

Proportions and relationships

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HES 3000-002, Sport Performance Analytics

Workflow

1. Open and inspect
2. Identify the observation and variables
3. Prepare the values
4. Choose and make a graph
5. Summarize
6. Interpret

What am I looking at?

- What does one row represent?
- What does each column represent?
- Which columns are numeric or categorical?
- Are there blank cells, labels mixed with values, or units in the cells?
- **Recall:** data characteristics slides.

Which values answer the question?

- Defined by your research question
 - Eg: study of offensive performance? → field goal percentage?
- **Recall:** unit of observation and data structure.

Inspecting data

game_id	team_id	opponent_team_id	athlete_id	field_goals_made	field_goals_attempted	fg%
401858379	201	38	5329888	0	2	0%
401851568	201	8	5329888	2	2	100%
401851564	201	245	5329888	3	4	75%
401851560	201	2579	5329888	0	0	0%
401808287	201	251	5329888	1	3	33%
401808280	201	142	5329888	2	5	40%
401808269	201	99	5329888	1	3	33%
401808262	201	2	5329888	0	0	0%
401808256	201	245	5329888	0	1	0%
401808247	201	2633	5329888	0	1	0%
401808239	201	61	5329888	3	3	100%
401808225	201	238	5329888	0	1	0%
401808216	201	96	5329888	2	4	50%
401808214	201	251	5329888	4	10	40%
401808206	201	8	5329888	1	1	100%
401808196	201	142	5329888	3	5	60%
401808191	201	2579	5329888	1	2	50%
401808184	201	333	5329888	2	3	67%
401808172	201	57	5329888	4	4	100%
401808166	201	245	5329888	1	1	100%
401808154	201	344	5329888	0	2	0%
401808151	201	145	5329888	0	1	0%
401826782	201	2400	5329888	2	3	67%
401858385	201	277	5144388	0	1	0%
401858385	201	277	5101844	3	3	100%
401858383	201	239	5144388	2	4	50%
401858383	201	239	5101844	2	2	100%

**sample rows from the full dataset*

1. Describing one row is a descriptive. Do not need to report metrics.
 - “each row represents a single athlete’s performance in one game of the 2026 season”
 - “a summary of a team’s performance last season”
2. Understanding the data is critical for all future steps.
 - Data type affects which summaries and graphs make sense. (mode vs mean, color vs position)
 - Is the data valid? (can fg% be 0 if fg attempted is 0?)

Which graph answers this question?

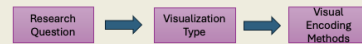
- *one numeric variable* → *distribution* → *histogram*.
- **Recall:** chart choice follows the question.

Choosing the Right Chart Type

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Match the chart to your question

- You would never do analysis before the question
- Same ideas apply here



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Comparisons (rank)

- Bar charts: Simple stat comparisons (points per game by player)
- Grouped/stacked bars: Multi-category comparisons (shooting % by zone)
- Dot plots: Clean alternative to bars, especially for rankings
- Radar/spider charts: Multi-dimensional player comparisons (use cautiously!)



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Distribution

- Histograms: Player height distribution, shot distance distribution
- Density plots: Smoothed shooting ranges
- Box plots: Comparing stat distributions across positions or teams
- Violin plots: Showing full distribution shape (sprint speeds by position)



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Relationships

- Scatter plots: Exploring correlations (height vs. rebounding)
- Correlation matrices: Multiple pairwise correlation
- Hexbin plots: scatter plots with aggregation (shot charts)



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Time series

- Line charts: Performance over season, in-game momentum
- Area charts: Cumulative stats over time



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Part-to-whole

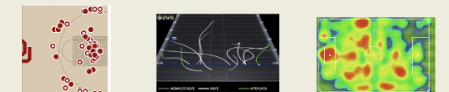
- Stacked bars: Composition of team offense (3PT vs. 2PT vs. FT)
- Tree maps: Allocating playing time or usage
- Pie charts: use cautiously



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Spatial/Location data

- Heat maps: Player positioning, shot locations
- Contour plots: Density of activity on field/court
- Field/court overlays: Context matters in sports



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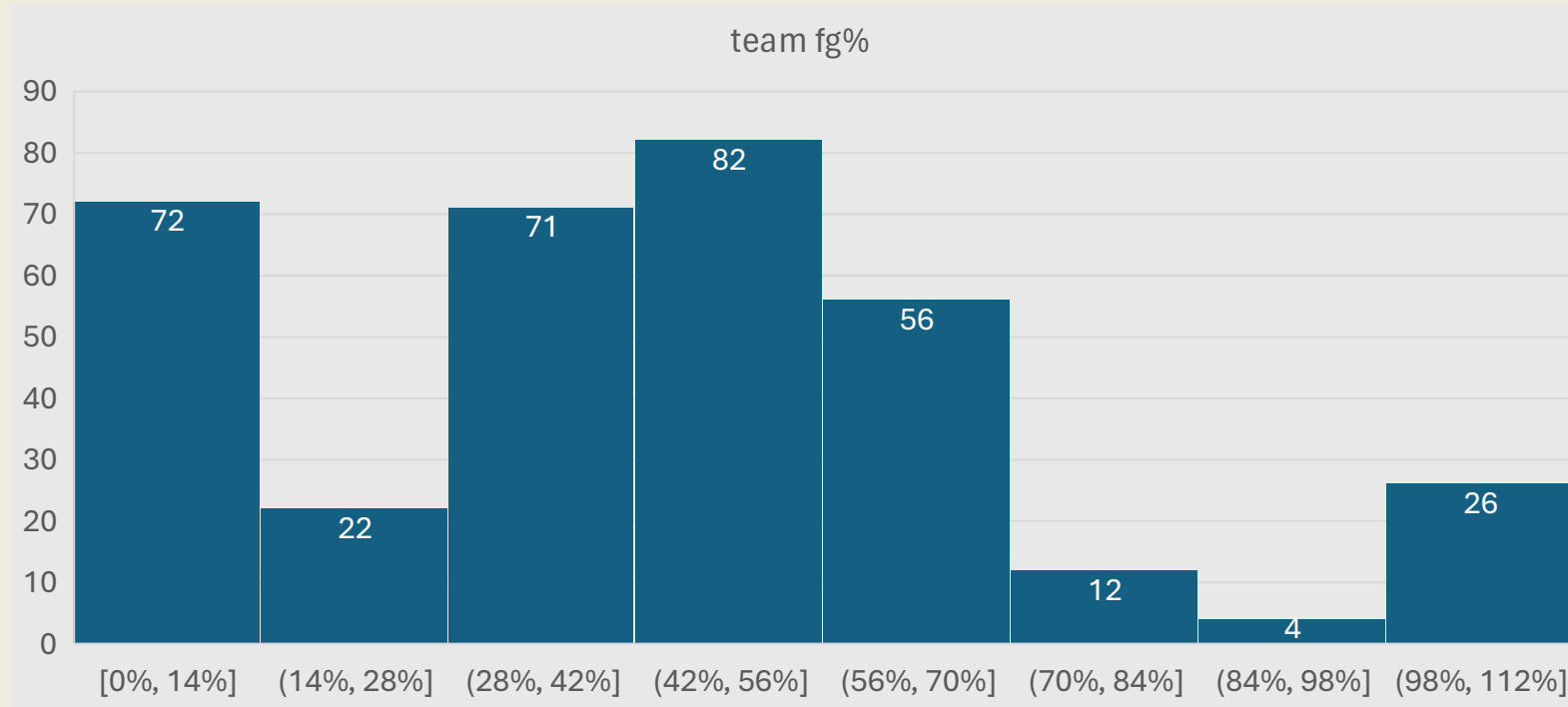
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Making the histogram

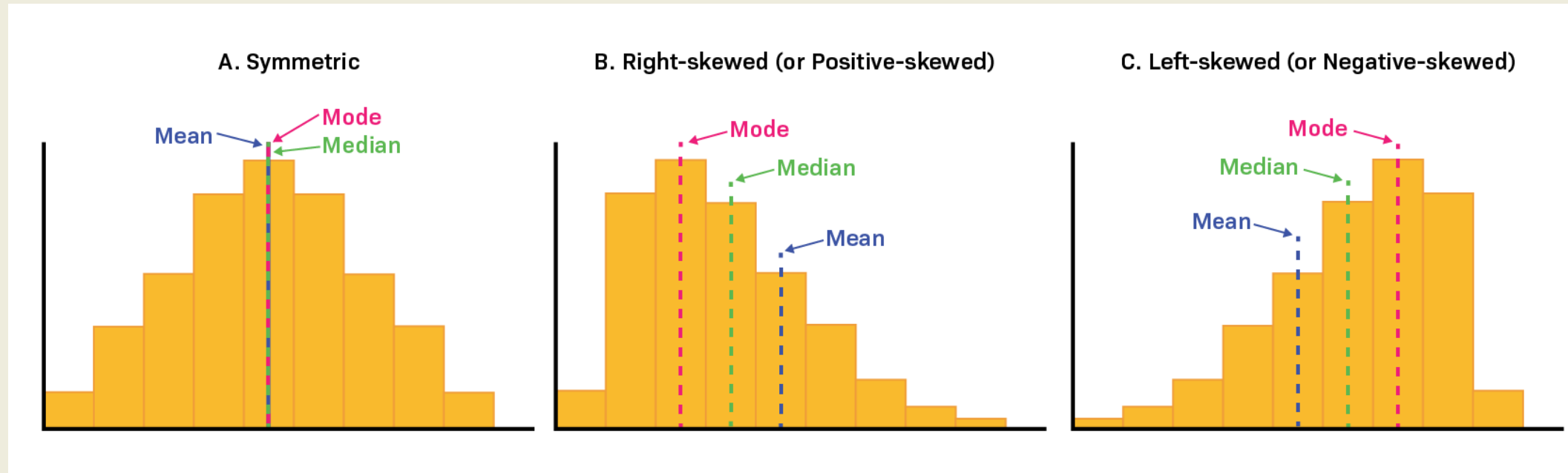
- one observation per row and values in one column
- **Recall:** long format data for visualization and analysis

What does the distribution tell us?



- Describe the shape, range/spread, clusters, unusual values.
- *Would the mean or median better describe a typical value here, and why?*

What does the distribution tell us?



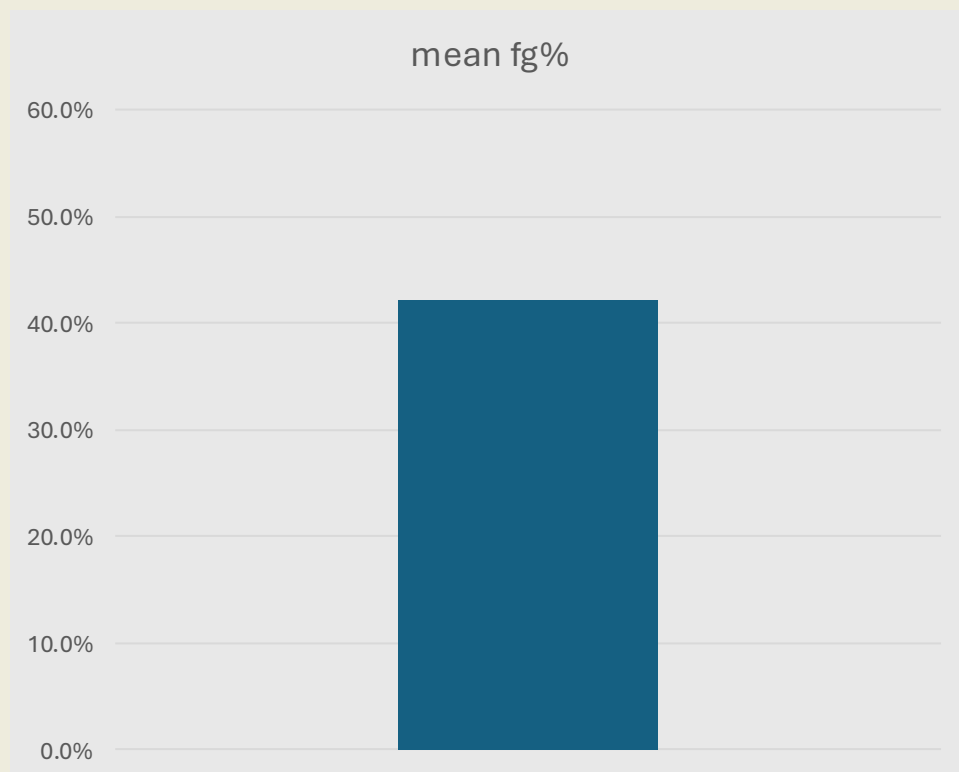
- **Recall:** center and spread lecture.

Bonus: What changes when we compare groups?

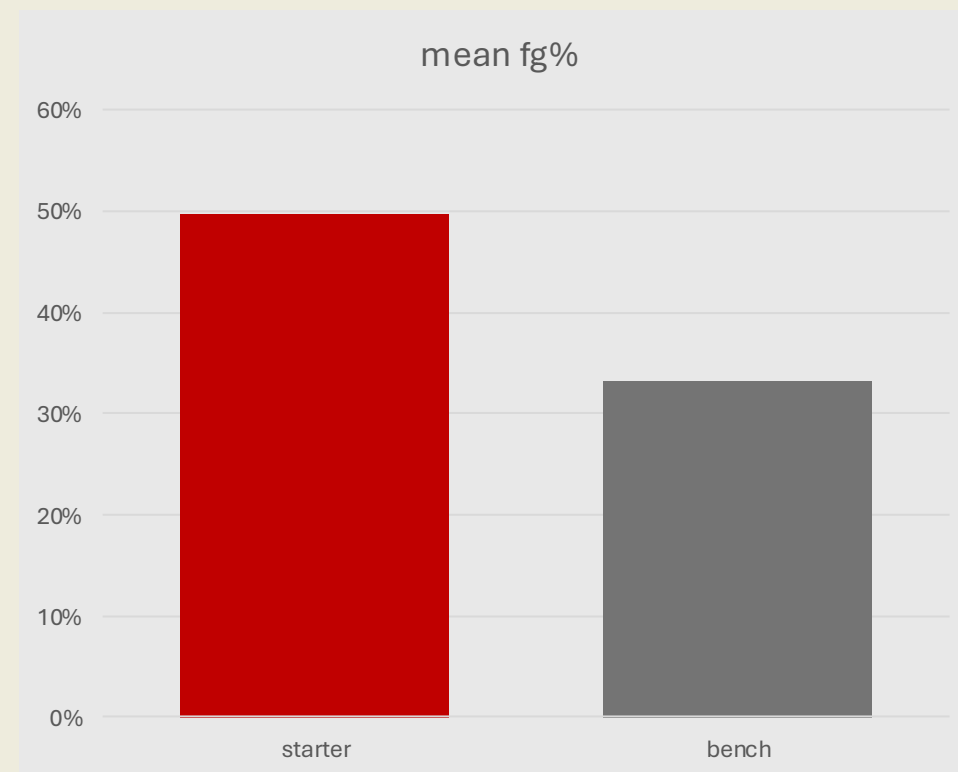
- Requires additional columns
- Do overall averages hides meaningful differences?

Bonus: What changes when we compare groups?

	n	mean fg%
Full team	345	42.1%



status	n	mean fg%
starter	185	50%
bench	160	33%



What does this tell us?

PROPORTIONS AND RELATIONSHIPS

Proportions in sports

Question:

- What proportion of recorded sessions were games, practices, or strength sessions?
- What proportion of successful shots were off short pass sequences (< 3 passes)?
- what proportion of the team improved in their agility test?

Proportions in sports

- A proportion describes a **part of a defined whole**.
- Proportion = number in a category ÷ total number
- Before calculating:
 - **What does one row represent?**
 - **What counts as the whole?**

	session_id ↕	session_type ↕	duration_min ↕	total_load ↕
1	1	Practice	60	280
2	2	Practice	75	360
3	3	Practice	80	410
4	4	Practice	65	310
5	5	Practice	90	450
6	6	Practice	70	330
7	7	Practice	85	420
8	8	Practice	55	260
9	9	Practice	95	470
10	10	Practice	75	390
11	11	Game	105	610
12	12	Game	115	700
13	13	Game	100	590
14	14	Game	120	740
15	15	Game	110	650
16	16	Game	95	560
17	17	Strength	40	170
18	18	Strength	50	210
19	19	Strength	45	190
20	20	Strength	55	230

Calculating proportions

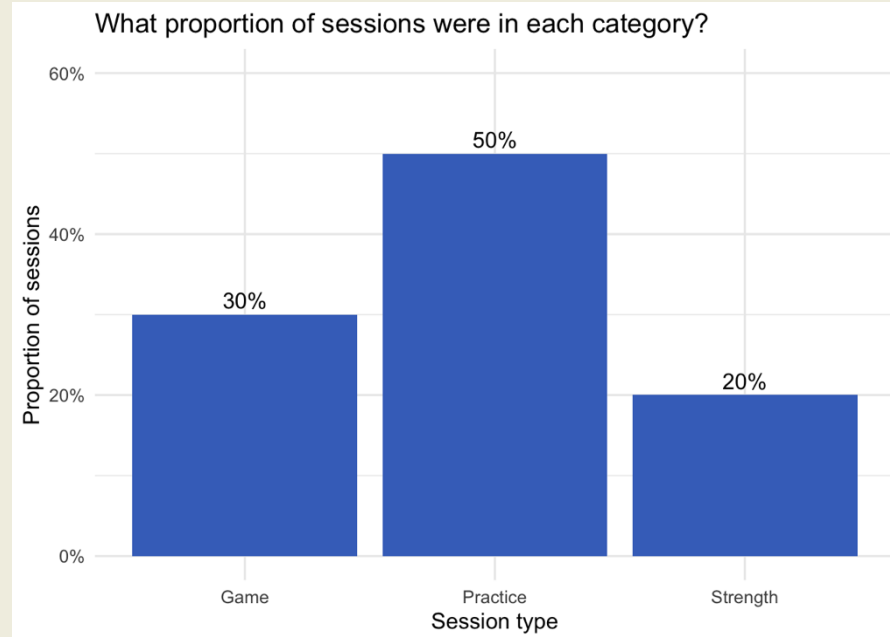
	session_id	session_type	duration_min	total_load
1	1	Practice	60	280
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19	19	Strength	45	190
20	20	Strength	55	230

	session_type	n	proportion
1	Game	6	0.3
2	Practice	10	0.5
3	Strength	4	0.2

Choosing a chart for proportions

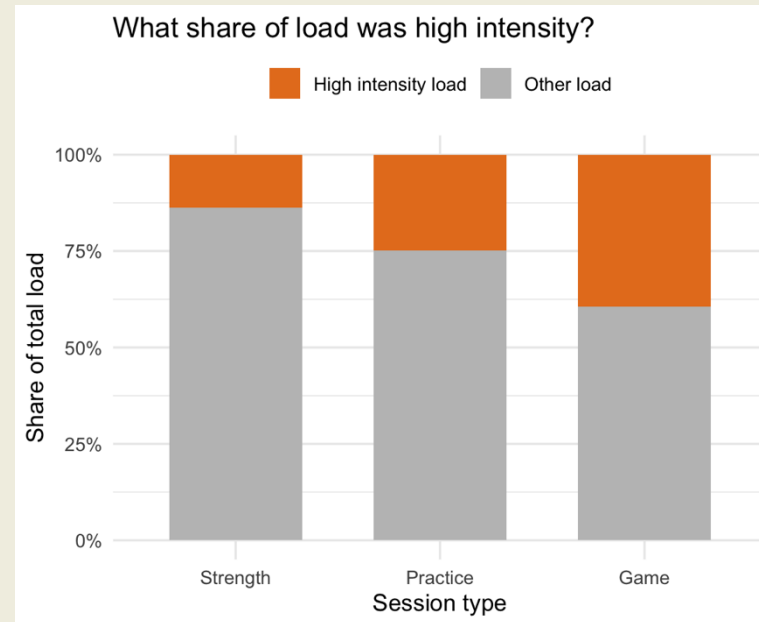
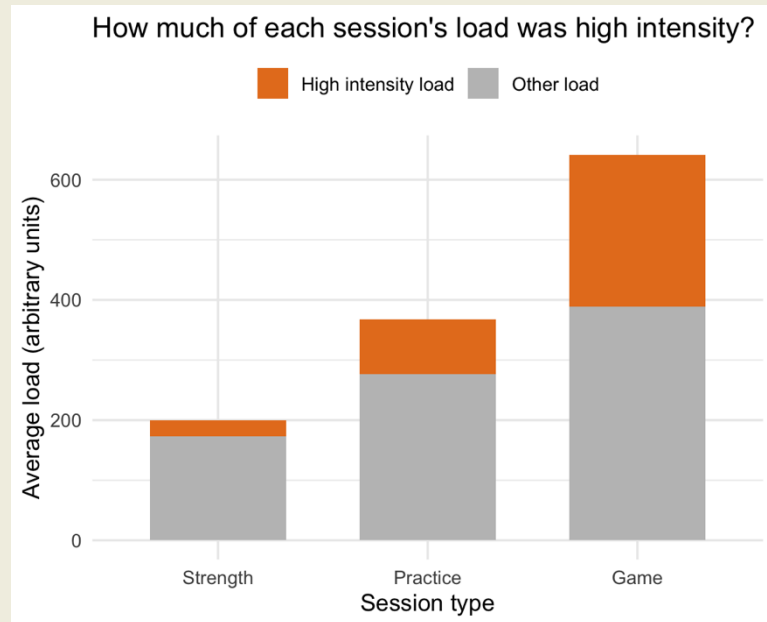
Question: Which session type was most common?

- Use a **bar chart** to compare categories.
 - categories on one axis and counts or percentages on the other.
 - Label whether the bars show **counts** or **percentages**.



Choosing a chart for proportions

Question: How is the load within each session type divided between high intensity and other load?

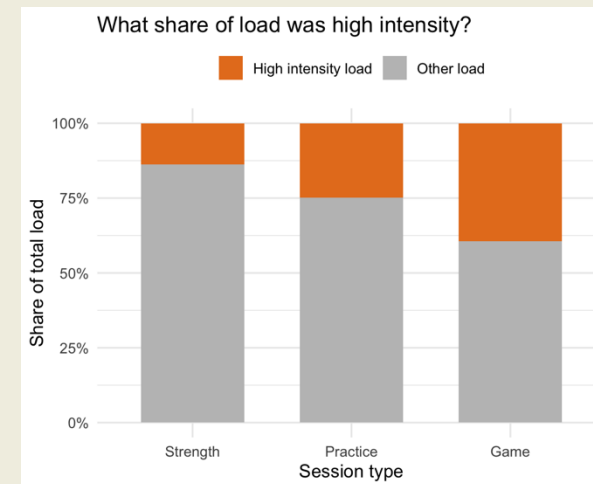
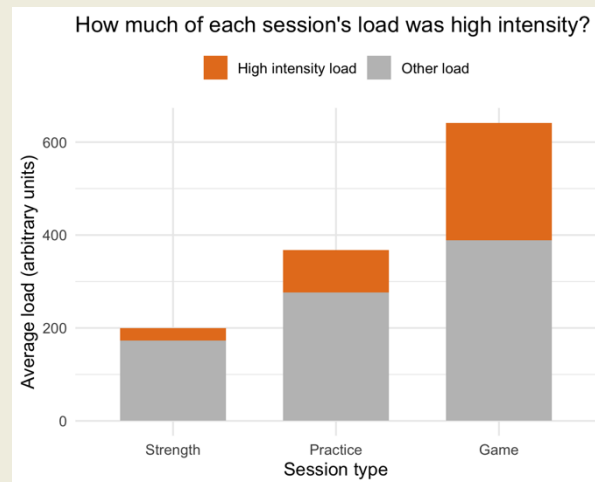


*Each bar represents the average total load per session. Each section show high intensity and other load.

Choosing a chart for proportions

Question: How is the load within each session type divided between high intensity and other load?

- **Stacked bar chart** when the parts add up to a meaningful total.
- **100% stacked bar chart** if the question is about the *proportion only*.



Data for relationships

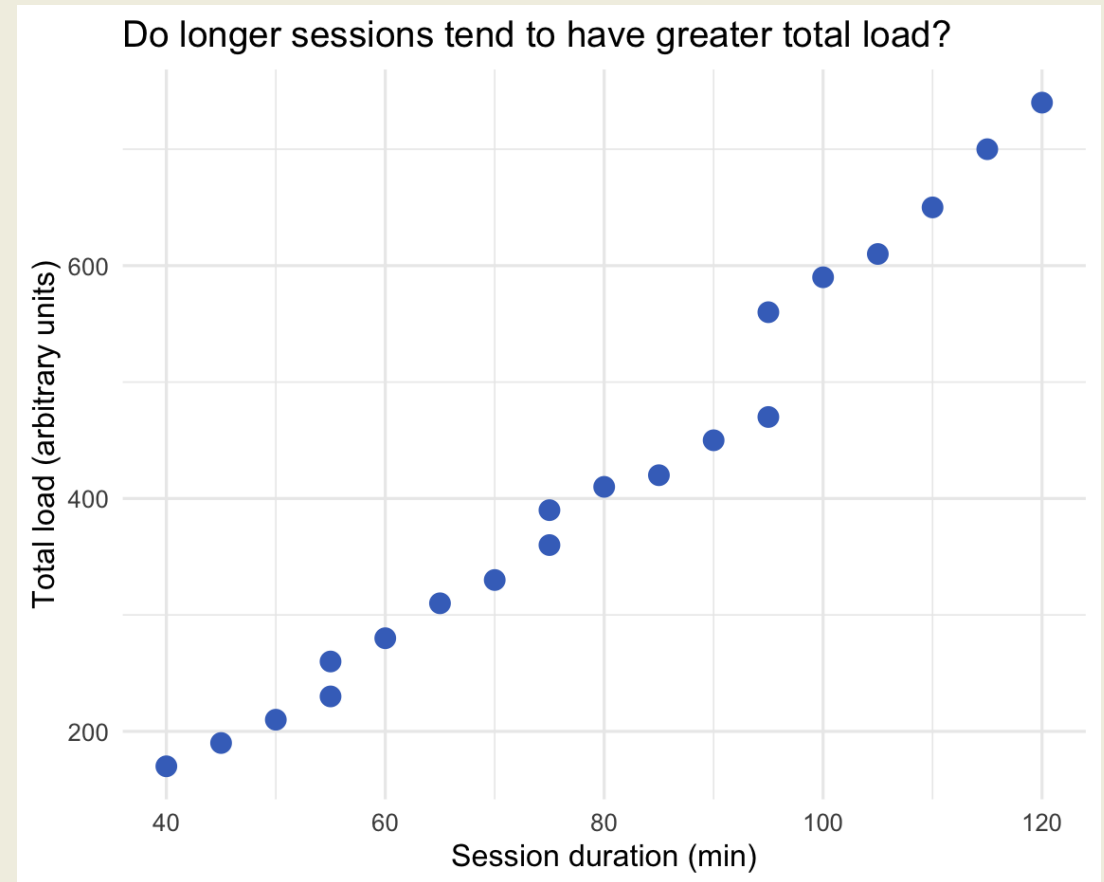
- We need two **numeric variables** measured on the **same observation**.

	session_id	session_type	duration_min	total_load
1	1	Practice	60	280
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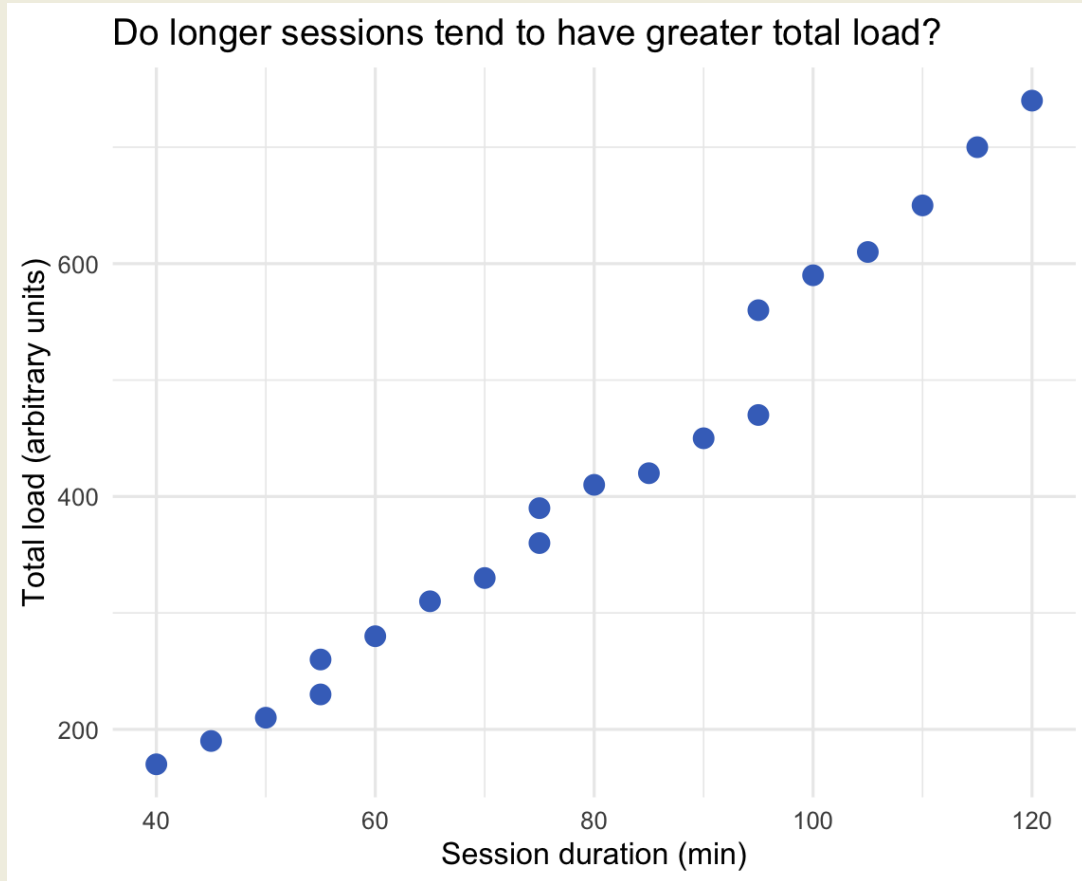
Visualizing relationships

Scatterplots:

- Explanatory variable on the x-axis and the outcome on the y-axis.
 - x-axis = duration
 - y-axis = total load
- Each point represents one session (or one athlete-session, depending on your data).



Reading a scatterplot



What pattern do you see?

- **Direction:**

- Do higher values of one variable tend to occur with higher or lower values of the other?

- **Shape:**

- Is the pattern roughly straight, curved, or unclear?

- **Spread:**

- How closely do the points follow the pattern?

- **Unusual points or groups:**

- Is anything worth a closer look?

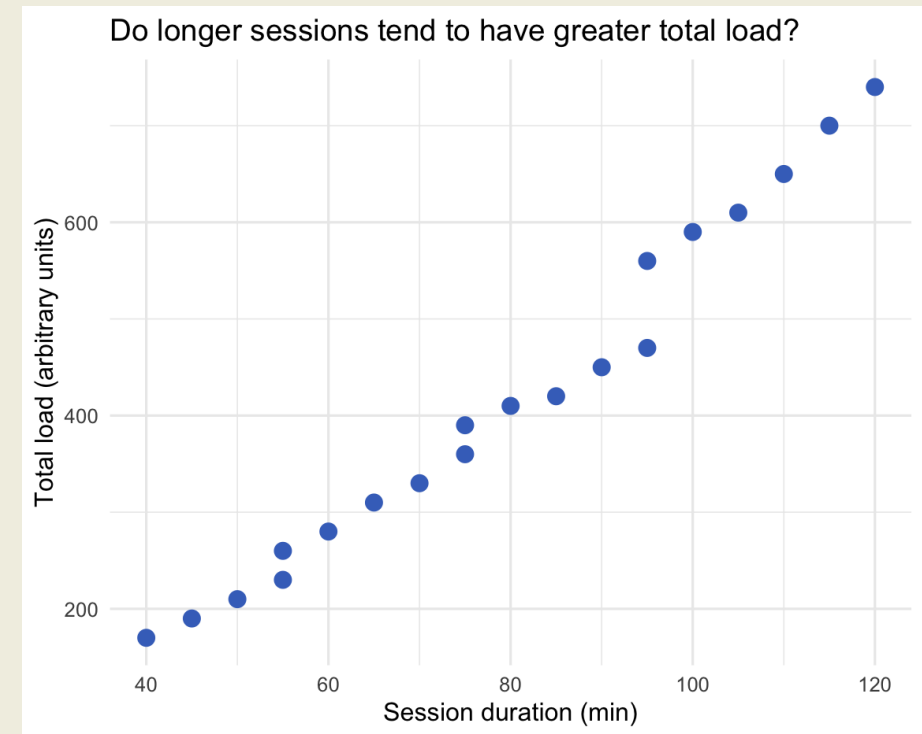
Interpreting graphs

Observation:

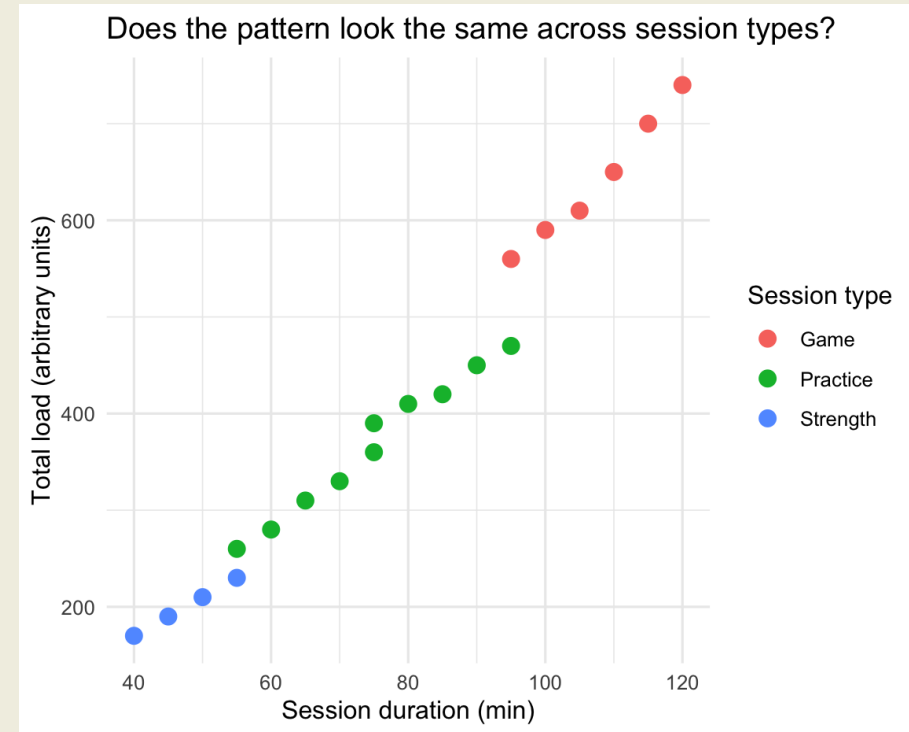
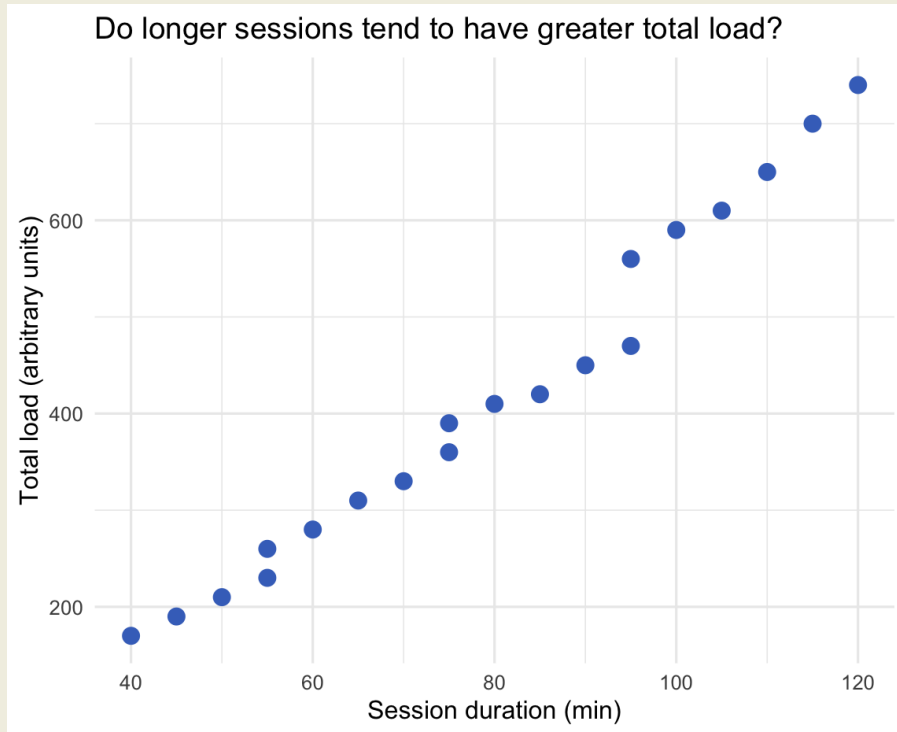
- Longer sessions tended to have greater total load.

Questions the graph alone cannot answer:

- Were those sessions more **intense**, or simply longer?
- Does duration itself explain the pattern?
- Would the same pattern appear within each session type?



Comparing groups



Summary

- Workflow: Start with the question, inspect the data, choose a graph, then interpret it.
- Proportions: Define the whole before calculating or graphing its parts.
- Relationships: Use a scatterplot to see how two numeric variables vary together. Describe the pattern before explaining it.

Next time

- Exam review: Bring questions from the lectures and case studies.